1. Scenario: You are working on a project that involves analyzing student performance data for a

class of 32 students. The data is stored in a NumPy array named student_scores, where each row

represents a student and each column represents a different subject. The subjects are arranged in the

following order: Math, Science, English, and History. Your task is to calculate the average score for

each subject and identify the subject with the highest average score.

Question: How would you use NumPy arrays to calculate the average score for each subject and

determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each

student in given order.

Code;

```
import numpy as np


# Marks of 4 students in the order:
# Math, Science, English, History
student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 88, 75],
    [70, 85, 80, 90],
    [85, 90, 82, 80]
])


# Calculate average score for each subject
subject_averages = np.mean(student_scores, axis=0)


# Subject names
```

```
subjects = ["Math", "Science", "English", "History"]

# Find the subject with the highest average
highest_index = np.argmax(subject_averages)
highest_subject = subjects[highest_index]
highest_average = subject_averages[highest_index]

# Display results
for i in range(4):
    print(subjects[i], "Average:", subject_averages[i])

print("\nSubject with highest average:", highest_subject)
print("Highest average score:", highest_average)
```

Output;

```
Math Average: 81.25
Science Average: 82.5
English Average: 83.75
History Average: 78.75

Subject with highest average: English
Highest average score: 83.75
```

2. Scenario: You are a data analyst working for a company that sells products online. You have

been tasked with analyzing the sales data for the past month. The data is stored in a NumPy array.

Question: How would you find the average price of all the products sold in the past month?

Assume 3x3 matrix with each row representing the sales for a different product

Code;

```
import numpy as np

# Sales data for 3 products
# Each row represents one product
# Columns represent different sales records/prices
```

```
sales_data = np.array([
    [100, 120, 150],
    [200, 180, 220],
    [80,  90,  110]
])

# Calculate the average price of all products sold
average_price = np.mean(sales_data)

# Display the result
print("Average price of all products:", average_price)
```

output;

```
Average price of all products: 138.88888888888889
```

3. Scenario: You are working on a project that involves analyzing a dataset containing information

about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a

NumPy array named house_data. Each row of the array represents a house, and the columns

contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of

houses with more than four bedrooms in the neighborhood?

Code;

```
import numpy as np

# House data
# Columns: Bedrooms, Square Feet, Sale Price
```

```
house_data = np.array([
   [3, 1200, 200000],
   [5, 1800, 350000],
   [4, 1500, 280000],
   [6, 2200, 450000],
   [5, 2000, 400000]
])

# Select houses with more than 4 bedrooms
selected_houses = house_data[house_data[:, 0] > 4]

# Get the sale prices of selected houses
sale_prices = selected_houses[:, 2]

# Calculate average sale price
average_price = np.mean(sale_prices)

print("Houses with more than 4 bedrooms:")
print(selected_houses)

print("\nAverage sale price:", average_price)
```

output;

```
============================================================
Houses with more than 4 bedrooms:
[[      5    1800 350000]
 [      6    2200 450000]
 [      5    2000 400000]]

Average sale price: 400000.0
>
============================================================
Houses with more than 4 bedrooms:
[[      5    1800 350000]
 [      6    2200 450000]
 [      5    2000 400000]]

Average sale price: 400000.0
>
```